

SAHITYA BISWAS

Ramakrishna Mission Vivekananda Educational and Research Institute , Belur

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My experience combines a strong foundation in machine learning and deep learning concepts with hands-on work in building end-to-end AI and data-driven systems. I am comfortable moving from understanding algorithms to implementing, training, and evaluating models in practical settings.

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute 2025 – 2027
*Master of Science in **Data Science and AI*** | Belur

Ramakrishna Mission Vivekananda Centenary College, Rahara 2022 – 2025
*Bachelors of Science in **Mathematics*** | CGPA : 7.56 Kolkata

PROJECTS

ScenGen | [RKMVERI](#) Jan-April . 2026

- **Brief Details** An LLM-guided Android GUI automation framework that combines computer vision, OCR, multi-agent reasoning, and ADB-based device control to perform intelligent scenario-based testing. The system dynamically understands mobile interfaces, identifies UI elements, plans actions using Large Language Models, and autonomously executes testing workflows without relying on predefined scripts or fixed UI locators.

• Gesture2Sentence 🗣️ | [RKMVERI](#) Feb.2026-April.2026

- **Brief Description:** An end-to-end Indian Sign Language recognition system using hierarchical deep learning models for large-scale isolated sign classification (2000 classes). Implemented video preprocessing and MediaPipe-based keypoint extraction pipelines for robust sign prediction from raw videos. Integrated a T5-based language generation module to convert predicted sign keywords into grammatically correct natural-language sentences. Conducted extensive model evaluation, inference optimization, and comparative experiments using CTRGCN architectures to improve recognition performance and scalability.

• Smart_CItY_Traffic_light_Control 🗣️ | [RKMVERI](#) Sep.2025-Dec.2025

- **Brief Description:** This project proposes a smart traffic light control system based on reinforcement learning that can adapt its decisions according to real-time traffic conditions at a four-way intersection. A lane-wise Q-learning approach is used, where the system observes the number of vehicles waiting in each direction and decides which signal should turn green. .

• JOB_NEXUS 🗣️ | [RKMVERI](#) Feb.2026-April.2026

- **Brief Description:** A distributed career recommendation platform leveraging Apache Spark Master-Worker architecture for parallel job and skill processing , combined with NLP-based skill extraction and semantic matching. Modeled career, skill, and course relationships using Neo4j graph databases and implemented real-time recommendation scoring with Cosine Similarity. Built an interactive glassmorphic dashboard featuring personalized career recommendations and a "What-If" career path simulator for skill-gap analysis and career planning.

Technical SKILLS

Languages: Python, R Programming, SQL, LaTeX

Machine Learning & AI : Machine Learning, Artificial Intelligence, Deep Learning, Natural Language Processing

Frameworks: OpenCV, Scikit-learn, Numpy, Pandas, Streamlit, Seaborn, PyTorch, Matplotlib.

Operating System: Windows, Linux(Ubuntu)

Tools: Git/GitHub, Ms Office .Ms Excel

ACHIEVEMENT/Certifications

- GATE 2025 (Data Science and Artificial Intelligence) – Qualified
- Bronze Honour – International Youth Math Challenge (IYMC) 2024